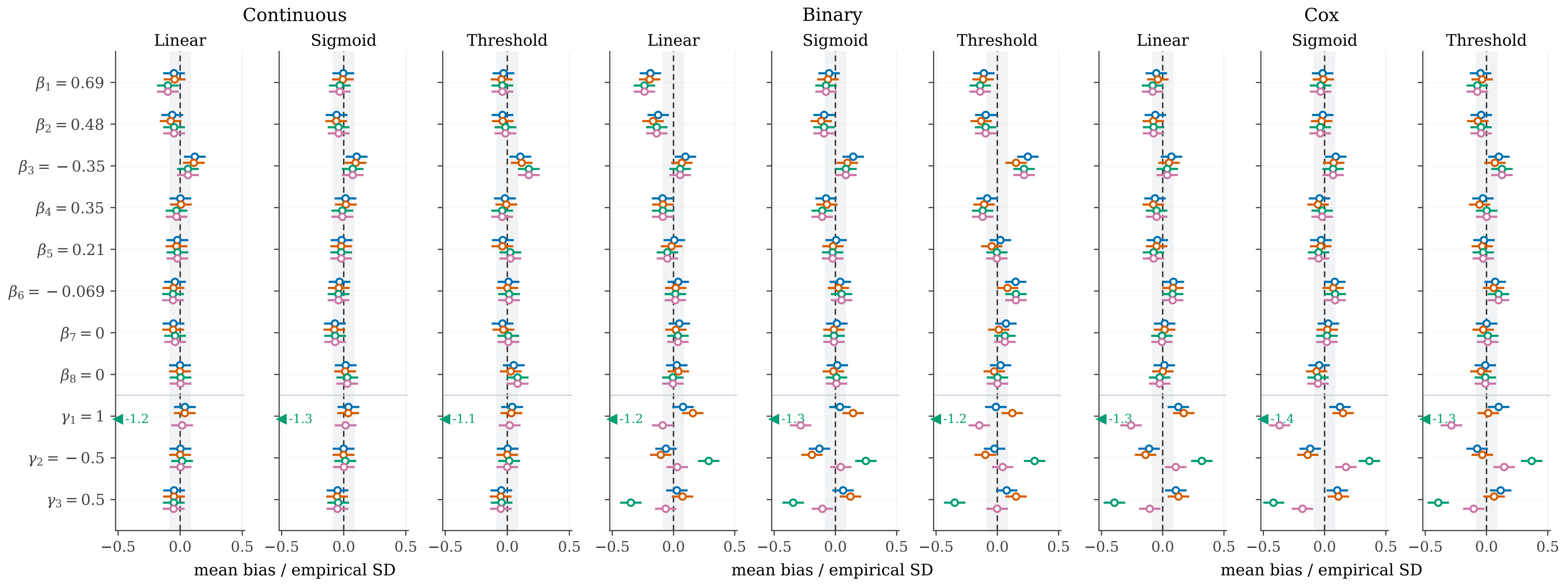
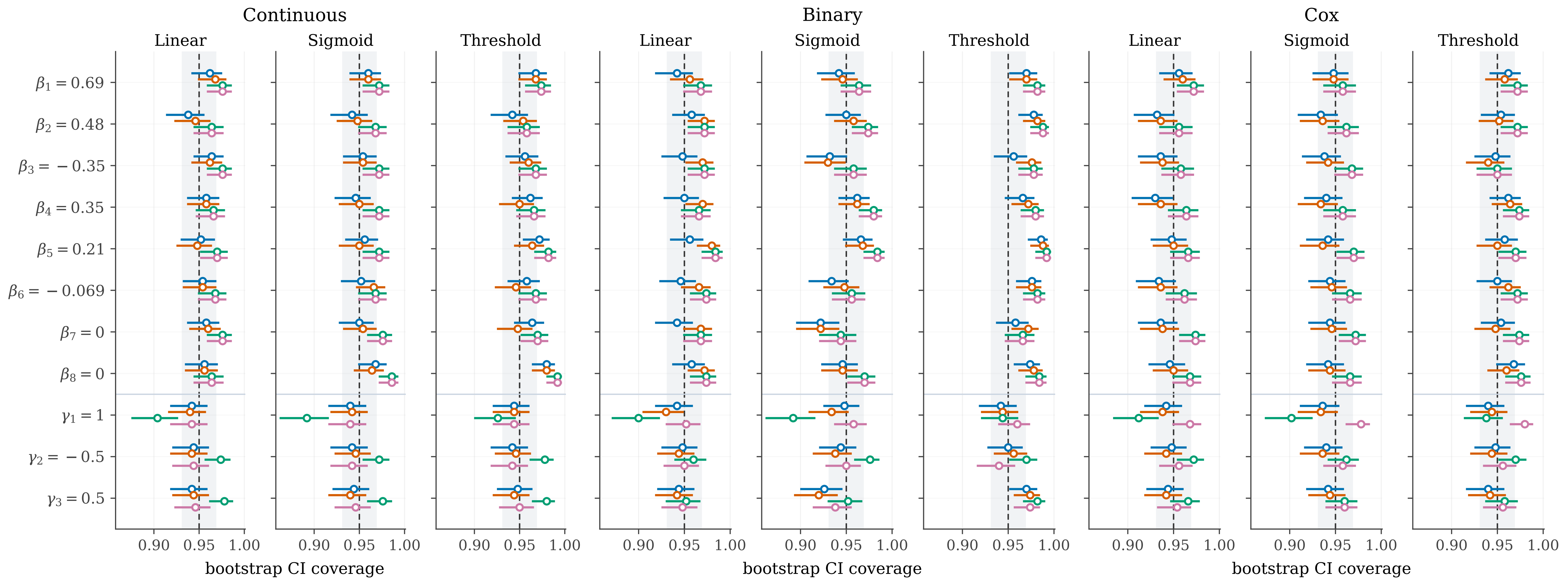
# Bootstrap inference in the simulations, n = 2000, 500 replicates per setting

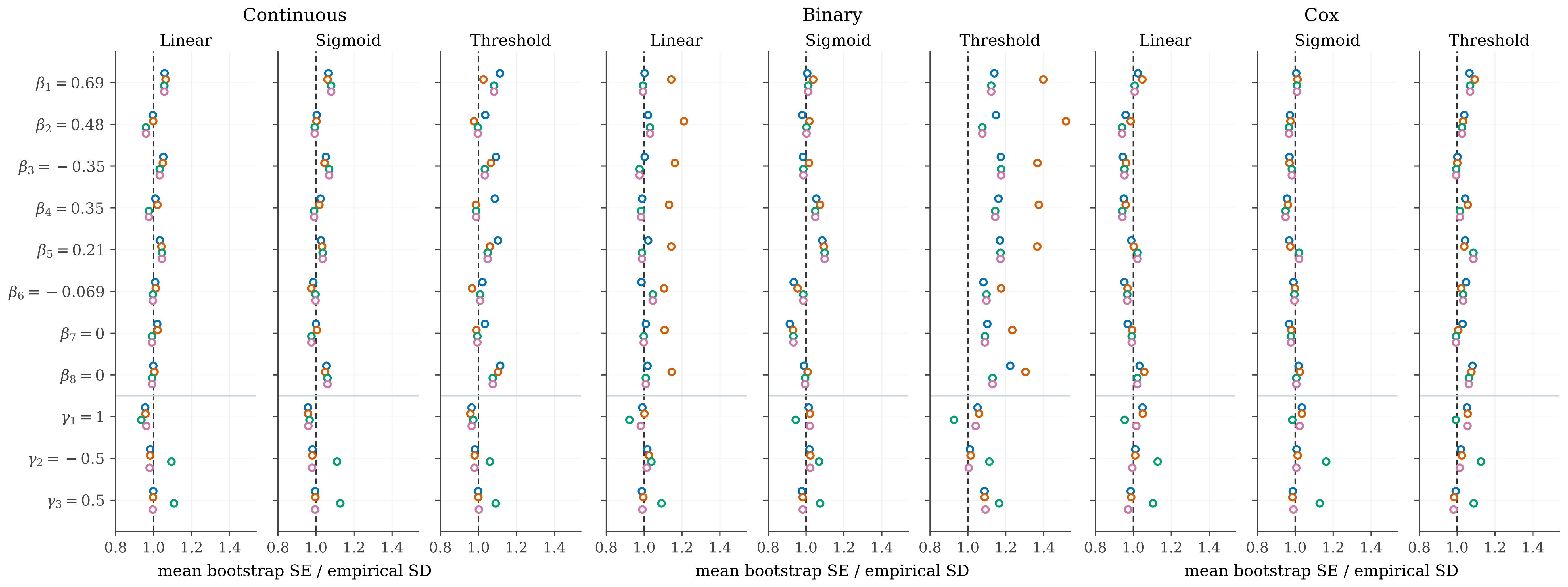
**A** Bias of each coordinate of $\beta$ and $\gamma$, in units of its empirical SD, with its 95% Monte-Carlo interval over the 500 replicates



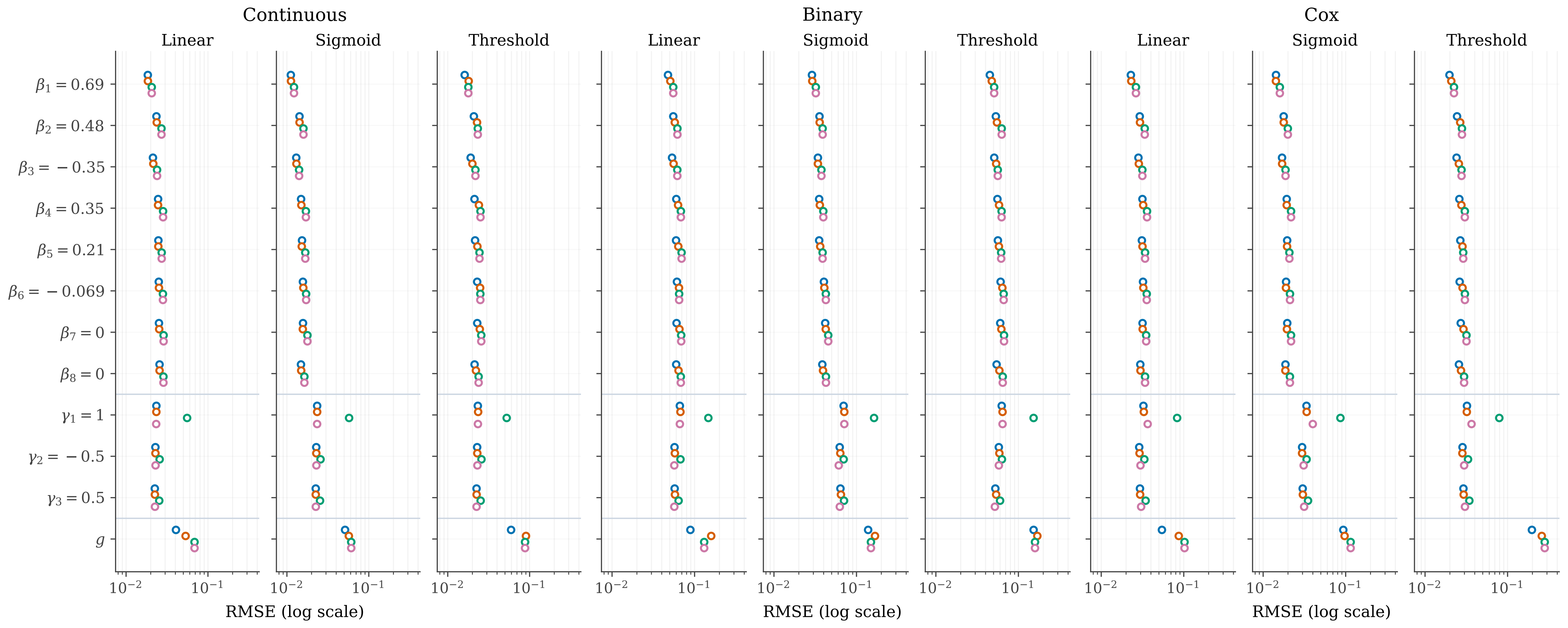
**B** Coverage of the 95% bootstrap CI for each coordinate, with its 95% Monte-Carlo (Wilson) interval



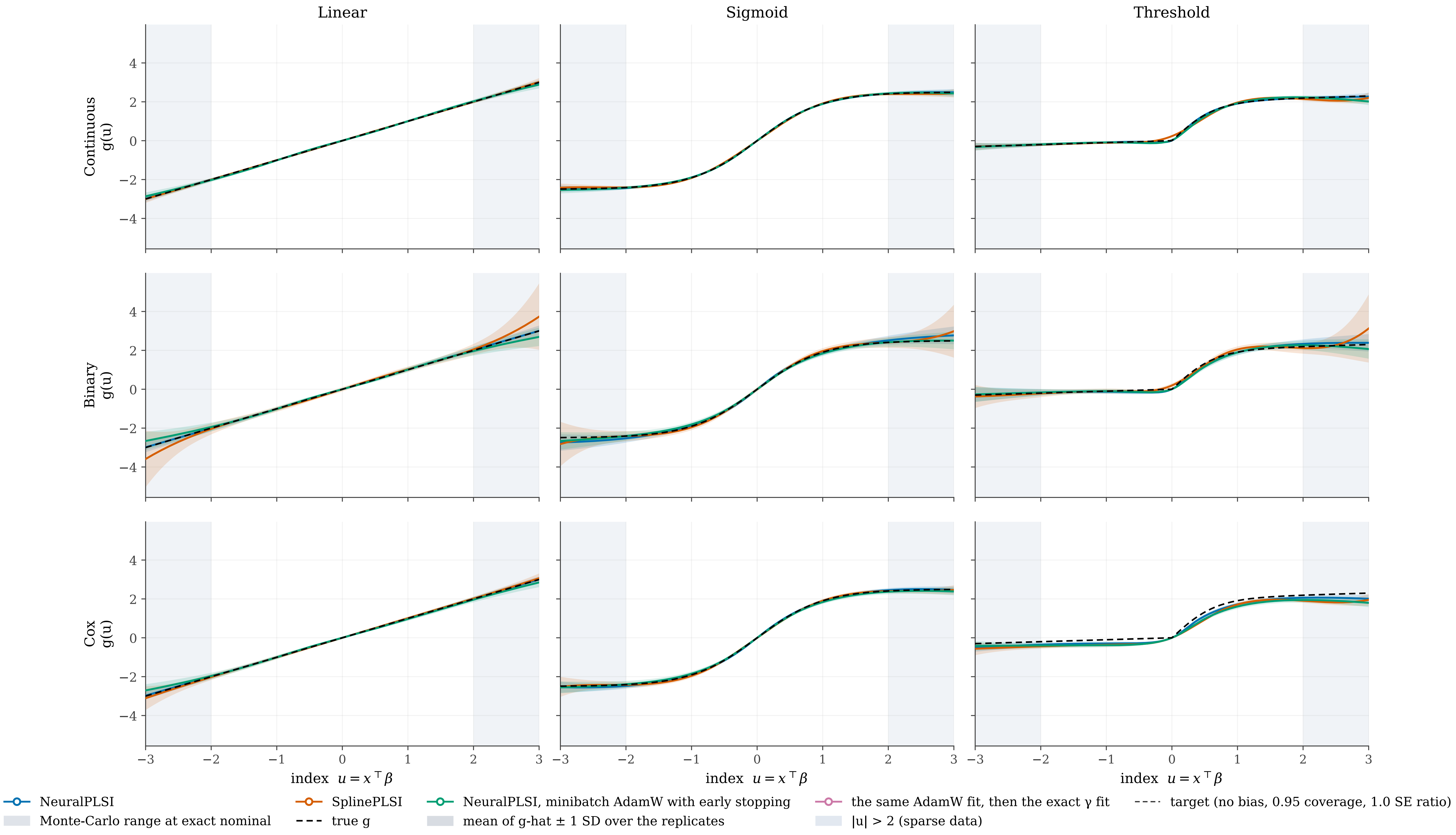
**C** SE calibration for each coordinate: below 1.0 means the intervals are too narrow



**D** Estimation accuracy: RMSE of each coordinate, and of g over |u| ≤ 2



**E** The estimated g: mean over the 500 replicates ± 1 SD, against the true g (NeuralPLSI-AdamGamma has the baseline's g and is not drawn)


Linear
Sigmoid
Threshold
Continuous g(u)
Binary g(u)
Cox g(u)
4
2
0
−2
−4
−3
−2
−1
0
1
2
3
index $u = x^\top \beta$
NeuralPLSI
Monte-Carlo range at exact nominal
SplinePLSI
true g
NeuralPLSI, minibatch AdamW with early stopping
mean of g-hat ± 1 SD over the replicates
the same AdamW fit, then the exact γ fit
|u| > 2 (sparse data)
target (no bias, 0.95 coverage, 1.0 SE ratio)